

Finite-cutoff coefficient-jet Wiener forest and complete-sector cancellation

TECT verification-first repository

claim A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION

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Purpose and scope

The coefficient-jet forest is classified exactly at finite cutoff, without closing the coefficient-jet model. For the covariance-matched six-real A1 production Gaussian, the complete non-aliased sharp-cube Bony forests reproduce the literal monomials. The summed value-derivative and double-cross contractions vanish because the full matrix $\mathbb{E}[X \otimes \partial_i X]$ is zero. For an uncentred coefficient pair, the only remaining lower-chaos Wick-conversion term is the finite tensor $\Sigma_\Lambda^{cd} Q_{\Lambda,i}^{ab}$.

This result explains how a fixed-sign cone-localized contraction can coexist with zero net complete-sector correction. It does not make an individual resonant jet convergent, does not improve the literal full product's spatial regularity, and does not identify a rational- B reconstruction with the A7 composite. A13 remains T4. The selected coefficient-jet gate is only partially advanced.

The scope is the fixed $L = 16$ three-torus, a centred stationary six-real finite Fourier Gaussian with the A1 production covariance, a common entrywise real-even cutoff, exact Fourier convolution with no intermediate projection, and Wick subtractions evaluated with the same cutoff law. A single deterministic root projection after the complete product is allowed. Node-dependent projections, circular aliasing, adapted shifts, continuum jet convergence, floor removal, and infinite volume are excluded.

1. Pinned Gaussian and parity identity

Write $V = L^3$, $\omega = 2\pi/L$, and

$$X_\Lambda(x) = V^{-1/2} \sum_{n \in \mathbb{Z}^3} R_\Lambda(n) \hat{X}(n) e^{i\omega n \cdot x}, \quad \mathbb{E}[\hat{X}^a(n) \hat{X}^b(m)] = \mathbf{1}_{n+m=0} C^{ab}(n). \quad (1.1)$$

Here $a, b \in \{1, \dots, 6\}$. The production complex symbol is a scalar even function of $|n|^2$ times the identity plus a real symmetric internal mass matrix. Its inverse is therefore entrywise real and even. The A7 six-real covariance is one half of its realification. With a common real-even R_Λ , every entry satisfies

$$C_\Lambda^{ab}(-n) = C_\Lambda^{ab}(n). \quad (1.2)$$

Let $Y_{\Lambda,i} = \partial_i X_\Lambda$. The complete mixed covariance matrix is

$$K_{\Lambda,i}^{ca} := \mathbb{E}[X_\Lambda^c(x) Y_{\Lambda,i}^a(x)] = \frac{i\omega}{V} \sum_n n_i C_\Lambda^{ca}(n) = 0. \quad (1.3)$$

The conclusion requires the matrix identity (1.3), not merely Hermiticity of the covariance. An odd imaginary off-diagonal covariance can be Hermitian and compatible with a real field while making $K_{\Lambda,i} \neq 0$.

Define the covariance-matched tensor

$$Q_{\Lambda,i}^{ab} =: Y_{\Lambda,i}^a Y_{\Lambda,i}^b := Y_{\Lambda,i}^a Y_{\Lambda,i}^b - \Gamma_{\Lambda,i}^{ab}, \quad \Gamma_{\Lambda,i}^{ab} = \mathbb{E}[Y_{\Lambda,i}^a Y_{\Lambda,i}^b]. \quad (1.4)$$

A subtraction from a different cutoff would leave an additional zero-chaos term and is not admitted.

2. Exact first-jet forest

The componentwise Wiener product formula gives

$$X^c Q_i^{ab} = : X^c Y_i^a Y_i^b : + K_i^{ca} Y_i^b + K_i^{cb} Y_i^a. \quad (2.1)$$

Thus the raw first jet has one third-chaos term and two first-chaos cross contractions. Under (1.3), its complete monomial is pure third chaos.

For a deterministic bilinear Fourier multiplier $m(p, r)$, the first-chaos part at output n is, with the convention $\partial_i \leftrightarrow i\omega n_i$,

$$(\mathcal{P}_1 J_1)^b(n) = \omega^2 n_i \sum_p p_i m(p, n-p) C^{ca}(p) X^b(n) + (a \leftrightarrow b). \quad (2.2)$$

The sign is positive because the contracted derivative mode is $-p$. When the complete multiplier sum is one, (2.2) reduces to (1.3) and vanishes. An individual resonant multiplier need not vanish.

3. Exact second-jet forest

Let $\Sigma_\Lambda^{cd} = \mathbb{E}[X_\Lambda^c X_\Lambda^d]$. For the Wick-centred coefficient pair,

$$\begin{aligned} : X^c X^d : Q_i^{ab} &= : X^c X^d Y_i^a Y_i^b : + K_i^{ca} : X^d Y_i^b : + K_i^{cb} : X^d Y_i^a : \\ &\quad + K_i^{da} : X^c Y_i^b : + K_i^{db} : X^c Y_i^a : + K_i^{ca} K_i^{db} + K_i^{cb} K_i^{da}. \end{aligned} \quad (3.1)$$

There are exactly four single-cross second-chaos families and two double-cross constants. They all vanish when the full matrix K_i vanishes.

For an uncentred pair,

$$X^c X^d Q_i^{ab} =: X^c X^d : Q_i^{ab} + \Sigma_\Lambda^{cd} Q_i^{ab} =: X^c X^d Y_i^a Y_i^b : + \Sigma_\Lambda^{cd} Q_i^{ab}. \quad (3.2)$$

Before the parity cancellation, the raw forest therefore has five second-chaos families and two zero-chaos families. If the inner XQ is first projected to pure third chaos, multiplication by the outer X has three second-chaos contractions and no double-cross constant before its own subtraction. These are distinct forest conventions:

$$2, \quad 4+2, \quad 5+2, \quad 3+0. \quad (3.3)$$

The scalar multiplicities are independently recovered by

$$\begin{aligned} H_1 H_2 &= H_3 + 2H_1, & H_2 H_2 &= H_4 + 4H_2 + 2H_0, \\ H_1^2 H_2 &= H_4 + 5H_2 + 2H_0, & H_1 H_3 &= H_4 + 3H_2. \end{aligned}$$

They verify the combinatorics but do not replace the six-real tensor formula.

In Fourier space, a scalar double cross at zero output contains

$$-\frac{\omega^2}{V^2} \sum_{p,q} p_i q_i w_\sigma(p, q, -p, -q) \left[C^{ca}(p) C^{db}(q) + C^{cb}(p) C^{da}(q) \right]. \quad (3.4)$$

The minus sign comes from the two Fourier derivatives, not from Wick's pairing rule. A positive cone can give a fixed negative contribution, as in the preceding A13 note. Equation (3.4) does not have a definite sign after all angular and Bony sectors are restored.

4. Complete sharp-cube reconstruction

Let $S_j = \mathbf{1}_{\{\|n\|_\infty \leq 2^j\}}$ and $\Delta_j = S_j - S_{j-1}$, with a declared low block. Every scale pair belongs to exactly one region

$$j \leq k - 2, \quad |j - k| \leq 1, \quad k \leq j - 2, \quad (4.1)$$

which defines the low-high, resonant, and high-low multipliers. Their sum is one on every Fourier pair. Deterministic multipliers commute with Wiener chaos projection, so summing the forest correctors over all three sectors recovers (2.1).

For the two second-jet parenthesisations, use the full nine-channel sums

$$\sum_{\alpha, \beta} \alpha(X^c, \beta(X^d, Q_i^{ab})), \quad \sum_{\alpha, \beta} \alpha(\beta(X^c, X^d), Q_i^{ab}). \quad (4.2)$$

Each sum separately equals the exact triple convolution. Their complete sums agree by associativity, but corresponding individual channels generally do not; their difference is an associator. Adding the two complete trees would double-count the monomial.

If X_J is supported through scale J , an XXQ output can reach the $J + 2$ block. Hence the resolution in (4.2) must cover the full convolution support. Intermediate projections such as $S_J[XS_J(XQ)]$ and $S_J[S_J(XX)Q]$ are not interchangeable. The theorem uses exact sparse convolution, or equivalently sufficient zero padding, followed at most by one common root projection.

Combining (1.3), (3.1), and (4.2) proves that the net complete-sector value-derivative and double-cross corrections vanish. This is compatible with the localized negative cone in (3.4): the omitted complementary sectors carry the opposite net coefficient. It does not license subtracting only the resonant correction while leaving the complementary sectors unchanged.

5. The finite Wick-conversion term

The production covariance behaves as $|n|^{-4}$. A three-dimensional shell contains $O(r^2)$ modes, so

$$\sum_n \|C(n)\| < \infty, \quad \Sigma_\Lambda \longrightarrow \Sigma. \quad (5.1)$$

Consequently $\Sigma_\Lambda^{cd} Q_{\Lambda, i}^{ab}$ in (3.2) is not a new divergent interaction counterterm at this monomial/full-partition level. It is nevertheless a load-bearing finite tensor Wick-conversion term. A later rational- B Taylor reconstruction must retain it and compare it with the exact A7 covariance-normal prescription.

The complete monomials in (2.1) and (3.2) retain the low-high spectrum of Q . The previous A13 analysis therefore still applies: literal full XQ and XXQ do not have the improved coefficient-jet regularities required by the translated-current model. The next proof must construct balanced or base-point-increment top-chaos jets, prove their coupled-cutoff convergence, and supply explicit compensating reconstruction terms.

6. Executed adversarial audit

The primary implementation independently derives the contraction counts, checks the three-way sharp-cube partition on every small-grid pair, and uses integer sparse convolution without intermediate truncation to reconstruct both nine-channel parenthesisations. It evaluates the production six-real covariance, verifies entrywise parity and $K_i = 0$, records a nonzero one-sided cutoff control, and confirms that each localized positive-cone coefficient is negative while the complete sector sum vanishes.

The non-importing implementation recovers every Hermite coefficient by Gauss–Hermite projection, reimplements the dyadic boundary rules, pairs n with $-n$ independently, and repeats the one-sided failure control. The integrated verifier pins all sources and requires 34/34 primary, 31/31 independent, and 17/17 integrated assertions, for 82/82 in total.

These finite fixtures are regression and falsifier evidence. They are not a cutoff-Cauchy estimate or a continuum regularity proof.

Devil’s-advocate

1. **VALID WITH MITIGATION: a real-even scalar regulator alone is enough for every matrix covariance.** It is not. The proof uses the entrywise-even A1 covariance, equivalently the complete matrix identity $K_i = 0$. An odd imaginary off-diagonal covariance is excluded.
2. **DISMISSED: the cone-localized logarithmic coefficient contradicts complete cancellation.** The cone is one sector and angular subdiagram. The exact complete partition adds complementary coefficients; only their sum is constrained by $K_i = 0$.
3. **VALID WITH MITIGATION: the two parenthesisations agree channel by channel.** They do not. The integer fixture records different channels with equal complete sums. Each tree must be reconstructed separately.
4. **UPHELD AGAINST AN INTERMEDIATE-PROJECTION READING: Galerkin projection may be inserted at every node.** Such projections break associativity and are outside this theorem. Only exact convolution and an optional common root projection are admitted.
5. **VALID WITH MITIGATION: $\Sigma_\Lambda Q_\Lambda$ can be dropped because it is finite.** Finiteness means it is not a new divergent counterterm. It remains part of the exact Wick conversion and must enter the A7 comparison.
6. **UPHELD AGAINST CLOSURE: the forest classification proves the coefficient-jet model.** It does not prove improved regularity, continuum convergence, rational coefficient reconstruction, adapted translation, or the Nelson estimate. The selected A13 gate remains open.

Result footer

Result ID:	A13-CLASSII-COEFFICIENT-JET-FOREST-FINITE-CUTOFF
Precise statement:	Complete non-aliased sharp-cube forests reproduce the covariance-matched finite-cutoff first and second coefficient monomials; summed value-derivative and double-cross corrections vanish, while raw XXQ keeps the finite tensor $\Sigma_\Lambda Q_\Lambda$.
Scope:	L=16; six-real A1 Gaussian; same-cutoff Wick ordering; entrywise-even common cutoff; exact convolution; no intermediate projection; finite cutoff only.
Dependencies:	A1-PRODUCTION-FUNCTIONAL-REALISATION; A7-CLASSII-RENORMALISED-ENERGY-COMPOSITE; A13 universal-Q subproof; A10-A13 sharp cubes.

Evidence grade: ANALYTIC, EXACT, EXECUTED.
Reproduction command: python codes/foundations/a13_classii_coefficient_jet_forest_verify.py
Expected output: 34/34 primary, 31/31 independent, 17/17 integrated;
aggregate 82/82; integrated PASS; exit 0.
Falsification gate: Any wrong forest count, incomplete Bony partition,
nonzero complete K or double-cross coefficient,
parenthesisation mismatch after full summation,
source-hash mismatch, or failed assertion.
Tier before / after: T4 / T4 (no tier change).
No-overclaim statement: No continuum jet convergence, improved full-product
regularity, A7 reconstruction, adapted shift, one-use
estimate, Nelson bound, measure, T5, T6, or T7.
Next required action: Prove coupled-cutoff convergence of balanced P3/P4
jets and the exact finite-term A7 reconstruction.